وزارة الاتصـــــــالات
وتكنولوجيا المعلومات

NTI
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الرواد الرقميون
DiGiLiANS

9-Month Diploma

Bidz: A Secure Web-Based Auction Platform with Real-Time Bidding and Transaction Control

Track: Software Development

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Agenda

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- Problem Definition
- Motivation & Objectives
- Implementation (Seller & Customer Flows)
- System Architecture
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Introduction

- **Bidz** is a web-based auction platform designed to provide a secure and structured environment for buying and selling products through online auctions.
- The platform allows users to participate as Customers or Sellers, while an Admin manages and monitors the entire platform through a dedicated dashboard.
- **Bidz** covers the complete auction lifecycle, from product verification and bidding to communication, transaction handling, and delivery confirmation.

Problem Definition

- Online auction platforms can face challenges that reduce trust between buyers and sellers, especially when dealing with valuable or diverse products.
- The main problems addressed by **Bidz** include:
  - Fake or inaccurate product information.
  - Insufficient product and document verification
  - Unsafe delivery confirmation
  - Lack of secure payment holding
  - Limited communication between buyers and sellers
  - Lack of an effective reporting and moderation mechanism
- These challenges may increase the risk of fraud, disputes, and unsuccessful transactions.

Motivation & Objectives

- **Bidz** was motivated by the need for a secure and flexible online auction platform where different types of products can be offered while maintaining product verification, transaction control, and communication between users.
- The limited availability of platforms combining these capabilities encouraged us to develop **Bidz**.
- **Bidz** aims to:
  - Provide a secure online auction environment.
  - Allow users to act as both Customers and Sellers.
  - Verify product information and legal documentation.
  - Support transparent and competitive bidding.
  - Enable real-time communication between bidders and sellers.
  - Provide reporting and administrative moderation.
  - Protect users and APIs through multiple security mechanisms.
  - Control transaction funds until delivery confirmation.
  - Automate auction expiration and management.

Implementation (1/2)

- **Bidz** implements a complete auction workflow.
- Seller Side:
  - Purchase an auction ticket with a duration from 1 to 31 days.
  - Submit the product details, images, and required legal documents.
  - Obtain product approval.
  - The auction becomes active only after both the Ticket and Approval requirements are completed.
  - Receive bids from customers.
  - Communicate individually with bidders using real-time chat.
  - Select a bid and close the deal.
  - Complete the delivery confirmation process.

Implementation (2/2)

- <u>Customer Side:</u>
- Browse available auctions.
  - Select a product.
  - Place a bid higher than the current bid.
  - Receive updates when another bidder places a higher offer.
  - Communicate with the seller.
  - Complete the transaction if the seller selects the bid.

Architecture (1/2)

- **Bidz** follows a Web Application architecture consisting of a React frontend, Node.js/Express backend, and MongoDB database.
- <u>Frontend Main Components:</u>
  - React
  - User and Admin interfaces
- <u>Backend Main Components:</u>
  - Node.js
  - Express.js
  - REST APIs
  - Business logic
  - Authentication & Authorization
  - Auction and transaction management

Architecture (2/2)

- Database Main Components:
  - MongoDB
  - Mongoose
- Real-Time Communication:
  - Socket.IO
- Security Layer:
  - JWT
  - HttpOnly Cookies
  - RBAC
  - Middleware
  - Input Validation
  - Sanitization
  - Rate Limiting
  - Helmet / CSP / CORS
  - File validation

Methodology

- The development of **Bidz** followed a structured software development process:
- Requirements Analysis: Identifying users, roles, business rules, auction lifecycle, and security requirements.
- System Design: Designing the application architecture, database models, authentication flow, and auction workflow.
- Development: Implementing the frontend, backend APIs, auction system, bidding mechanism, real-time chat, wallet flow, and administrative dashboard.
- Security Integration: Applying authentication, authorization, input validation, sanitization, rate limiting, secure cookies, and file upload restrictions.
- Testing: Testing APIs and application functionality using Postman and manual testing.

Results

Bidz successfully implements an integrated auction workflow that connects:

- Product submission and verification
- Auction ticket management
- Automated and administrative approval
- Competitive bidding
- Real-time bidder–seller communication
- Reporting and moderation
- Controlled transaction fund handling
- Delivery confirmation
- Administrative platform management
- The system also integrates multiple security mechanisms to protect authentication, APIs, user input, and uploaded files.
- The implemented prototype demonstrates the feasibility of providing a secure, flexible, and controlled online auction experience.

Future Work

- Future development of **Bidz** could include:
- Integration with real online payment gateways.
- Advanced fraud and suspicious activity detection.
- Enhanced product and seller verification.
- AI-based product and auction recommendations.
- Integrated delivery services.
- Mobile application.
- Advanced analytics and auction insights.
- Enhanced dispute resolution.
- Rating and review system.
- These improvements could further enhance the platform's security, scalability, trust, and user experience.

References

- React Documentation. React.dev.
- Node.js Documentation. Node.js Foundation.
- Express.js Documentation. OpenJS Foundation.
- MongoDB Documentation. MongoDB, Inc.
- Mongoose Documentation.
- Socket.IO Documentation.
- JWT.io. JSON Web Tokens Introduction.
- OWASP Foundation. OWASP Top 10 Web Application Security Risks.
- OWASP Foundation. OWASP Cheat Sheet Series.
- Helmet.js Documentation.
- Postman Documentation. Postman, Inc.

Thank You